Matrix Size: $N = 512$

Task 1

Assuming s represents the fraction of zeros (sparsity).

(a) Dense MVM compute (FLOPs):
Expression: $2 \times N^2$
Calculated: $2 \times 512^2 = 524{,}288$ FLOPs

(b) Dense memory bytes:
Expression: $4 \times N^2$
Calculated: $4 \times 512^2 = 1{,}048{,}576$ Bytes

(c) Sparse compute (FLOPs):
Expression: $2 \times N^2 \times (1 - s)$
Calculated: $524{,}288 \times (1 - s)$ FLOPs

(d) Sparse memory bytes (CSR Format):
Expression: (4 bytes × nnz) + (4 bytes × nnz) + (4 bytes × (N + 1))
Formula: $8 \times N^2 \times (1 - s) + 4 \times (N + 1)$
Calculated: $8 \times 262{,}144 \times (1 - s) + 4 \times 513 = 2{,}097{,}152 \times (1 - s) + 2{,}052$ Bytes

Task 2

The theoretical speedup of using the sparse format versus the dense format is the ratio of their compute requirements.
Speedup Formula: Dense FLOPs / Sparse FLOPs = $(2 \times N^2) / (2 \times N^2 \times (1 - s)) = 1 / (1 - s)$
Sparsity for 2x Speedup:
$2 = 1 / (1 - s)$
$1 - s = 0.5$
$s = 0.5$ (or 50%)

Task 3

The breakeven point occurs when the memory required to store the CSR metadata equals the memory required for a dense matrix.
Equation: $4 \times N^2 = 8 \times N^2 \times (1 - s) + 4 \times (N + 1)$
Solving for s:
$1{,}048{,}576 = 2{,}097{,}152 \times (1 - s) + 2{,}052$
$1{,}046{,}524 = 2{,}097{,}152 \times (1 - s)$
$1 - s = 0.49902$
$s = 0.50098$ (or 50.1%)

Task 4

Assuming the execution is entirely bound by a memory bandwidth of 320 GB/s, evaluate the speedup at 90% sparsity ($s = 0.9$).
Dense Execution Time (T_dense):
T_dense = 1,048,576 Bytes / (320×10^9 Bytes/s) = 3.2768 µs
Sparse Memory at $s = 0.9$:
Sparse Bytes = $2{,}097{,}152 \times (1 - 0.9) + 2{,}052 = 209{,}715.2 + 2{,}052 = 211{,}767.2$ Bytes
Sparse Execution Time (T_sparse):
T_sparse = 211,767.2 Bytes / (320×10^9 Bytes/s) = 0.66177 µs
Final Speedup:
Speedup = T_dense / T_sparse = 3.2768 µs / 0.66177 µs = 4.95

At 90% sparsity (s=0.9), your hardware achieves a 4.95x end-to-end execution time speedup compared to processing the dense matrix.